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Normal Force Acting on a Hemispherical Nose Tip

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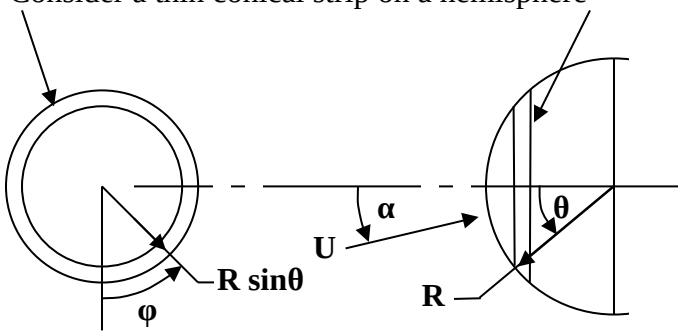
by C.P.Hoult

Introduction

Blunted, hemispherical nose tips are commonly used on sounding rockets. So, it behooves the analyst to develop approximate models for estimating their contribution to the static stability of a rocket. We start from known solutions for the incompressible and hypersonic flows over a sphere. The former is just that for a doublet plus free stream, and the later is based on the Lee modification to Newtonian flow.

Analysis

Consider a thin conical strip on a hemisphere



Any pressure on an element of this strip will contribute an element to its normal force, **dN**:

$$\mathbf{dN = p * (R\,d\theta * R\sin\theta\,d\varphi) * \cos\varphi * \sin\theta}, \text{ or}$$

$$\mathbf{dN = (q*C_p + p_\infty) * R^2 * \sin^2\theta\,d\theta * \cos\varphi\,d\varphi}.$$

Here **q** = Dynamic pressure,
C_p = Pressure coefficient, and
p_∞ = Ambient atmospheric pressure.
The remaining variables are defined in the sketches above.
Next, the pitching moment **M** about the nose tip is

$$\mathbf{dM = R\,(1 - \cos\theta)\,dN}$$

Then, the center of pressure for a hemispherical nose **xcp** is located a distance behind its tip of

$$\mathbf{xcp = M / N}.$$

Now, ref. 1 shows that the pressure distribution on a sphere in incompressible flow is

$$C_p = 1 - (9/4)\sin^2\theta$$

as long as $\alpha = 0$. The corresponding result for supersonic flow is presented in ref. 2:

$$C_p = C_{p0}\cos^2\theta,$$

where C_{p0} = the stagnation point pressure coefficient obtained behind a normal shock followed by isentropic compression.

The approach taken is to assume that if there is a small angle of attack α both pressure distributions would follow the above relationships, but with a slightly different θ measured from the stagnation point. A little spherical trigonometry shows that, for small α, the modified $\theta = \theta'$ is

$$\theta' = \theta - \alpha\cos\varphi.$$

Then an element of normal force **N** in incompressible flow is given by

$$dN = q\,(1 - (9/4)\sin^2\theta')\,R^2\sin^2\theta\,d\theta\cos\varphi\,d\varphi.$$

Retaining only terms in α gives

$$dN = (9/2)\,\alpha\,q\,R^2\sin^3\theta\cos\theta\,d\theta\cos^2\varphi\,d\varphi.$$

Integrating φ from 0 to 2π and θ from 0 to θ_T = the value of θ when the hemisphere becomes tangent to a conical frustum gives

$$N = (9\,\pi/8)\,\alpha\,q\,R^2\sin^4\theta_T.$$

The normal force coefficient slope is then

$$C_{N\alpha} = (9\,\pi/8)\,R^2\sin^4\theta_T / S_{REF},$$

where S_{REF} = the aerodynamic reference area, and

$C_{N\alpha}$ = the normal force coefficient slope.

Consider the case of a full hemisphere nose for which $\theta_T = \pi/2$. If the reference area is the body cross section area

$$C_{N\alpha} = 9/8.$$

This is well below the slender body value of 2. The critical Mach number (the lowest for which sonic flow is first observed locally) for a sphere is about 0.41. Thus, as long as the

burnout Mach number is less than 0.41, there will be no wave drag penalty for a hemispherical nose while there will be significant mitigation of any static stability issues. This result is plotted in Fig. 1 below:

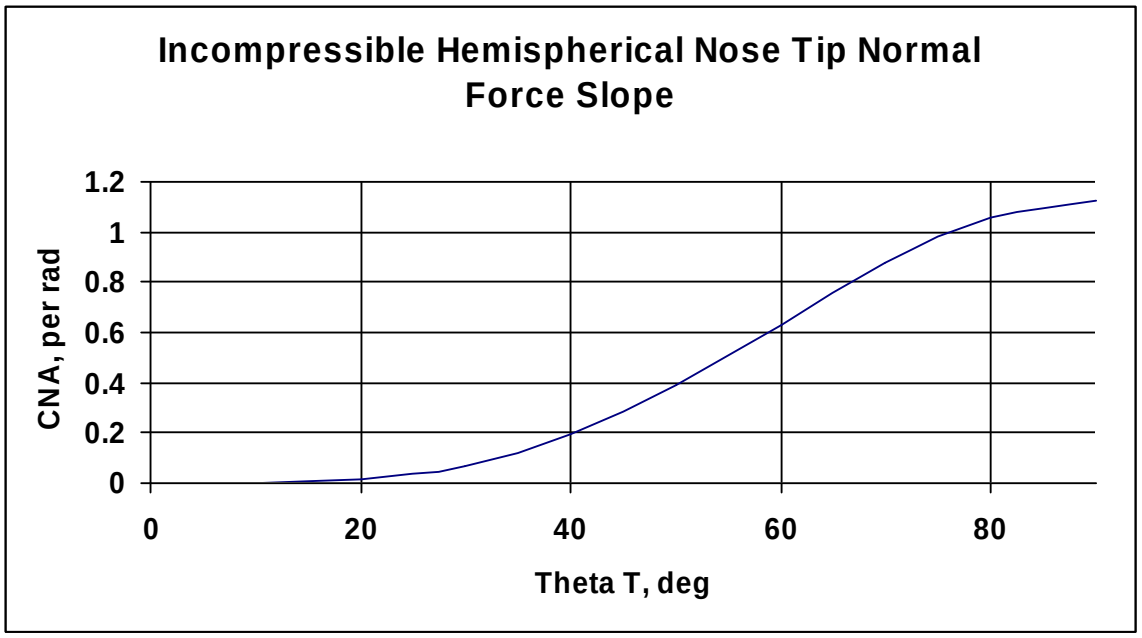


Figure 1

The corresponding pitching moment is

$$dM = R\,dN - (9/2)\,\alpha\, q\, R^3 \sin^3\theta \cos^2\theta\, d\theta \cos^2\varphi\, d\varphi,$$ or

$$M / \alpha\, q\, R^3 = (9\,\pi/8) \sin^4\theta_T - (9\,\pi/2) \left[(1/5) \cos\theta_T \sin^4\theta_T - (1/15) \cos\theta_T (\sin^2\theta_T + 2) + 2/15\right]$$

With this, the hemispherical nose center of pressure is

$$xcp/R = \left\{\sin^4\theta_T - 4 \left[(1/5) \cos\theta_T \sin^4\theta_T - (1/15) \cos\theta_T (\sin^2\theta_T + 2) + 2/15\right]\right\} / \sin^4\theta_T$$

This result is plotted in Fig. 2 below. Interestingly, it will be shown that the supersonic (modified Newtonian flow) result is exactly the same.

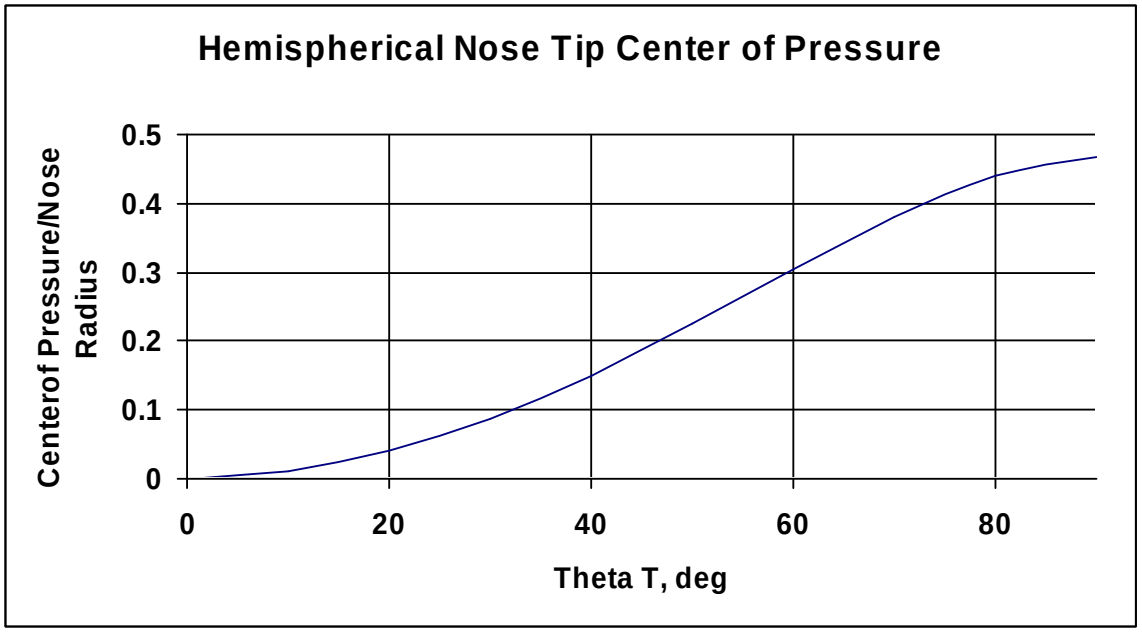


Figure 2

The supersonic case proceeds along similar lines. Again, retaining only first order terms in **α** results in

$$dN = 2\,C_{p0}\,\alpha\,q\,R^2\sin^3\theta\cos\theta\,d\theta\cos^2\varphi\,d\varphi.$$

The normal force coefficient slope is then

$$C_{N\alpha} = (\pi/2)\,C_{p0}\,R^2\sin^4\theta_T / S_{REF}.$$

For a full hemisphere $C_{N\alpha}$ based on body cross section area is,

$$C_{N\alpha} = (1/2)\,C_{p0}.$$

This, too, is less than the supersonic normal force developed by a slender cone. However, at supersonic speeds a hemispherical nose tip has far more wave drag than a slender cone. Note that the supersonic $C_{N\alpha}$ has the same form as that for incompressible flow.

The pitching moment now becomes

$$dM = R\,dN - 2\,C_{p0}\,\alpha\,q\,R^3\sin^3\theta\cos^2\theta\,d\theta\cos^2\varphi\,d\varphi,$$ or

$$M / \alpha\,q\,R^3 = (\pi/2)\,C_{p0}\sin^4\theta_T - 2\,C_{p0}\left[(1/5)\cos\theta_T\sin^4\theta_T - (1/15)\cos\theta_T(\sin^2\theta_T + 2) + 2/15\right].$$

The supersonic nose tip center of pressure is found as before:

$$xcp/R = \{\sin^4\theta_T - 4\,[(1/5)\cos\theta_T\sin^4\theta_T - (1/15)\cos\theta_T(\sin^2\theta_T + 2) + 2/15]\} / \sin^4\theta_T.$$

Interesting! The hemispherical nose tip center of pressure is independent of Mach number. Whodathunk that?

A final issue for the supersonic case is how to patch the Newtonian flow over a hemisphere into the shock expansion model used for the remainder of the body of revolution. Clearly the shock layer pressure at the nose base is

$$C_p = C_{p0}\cos^2\theta_T.$$

The classical Newtonian theory states that the shock layer would be very thin, and consequently, two dimensional oblique shock relations will be approximately valid. The primary concern arises when the local slope is small and/or the Mach number is low supersonic. In such a caser, the shock layer is not thin, but thick. Reference 3 provides the tools to resolve this dilemma. First, eq.(128) of ref.3 can be solved for the shock angle β that meets the other conditions. First,

$$p = q^*C_p + p_\infty, \text{or}$$

$$p/p_\infty = (\gamma/2)\,M\infty^2\,C_p + 1.$$

Next, β can be found from eq.(128):

$$\sin\beta = \sqrt{[(\gamma + 1)\,p/p_\infty + (\gamma - 1)] / 2\,\gamma\,M\infty^2}.$$

Here, $M\infty$ = The free stream Mach number, and

γ = The ratio of specific heats for air ≈ 1.4.

Knowing $\sin\beta$, eq.(132) of ref. 3 can then be used to find the shock layer Mach number **M:**

$$M^2 = [(\gamma + 1)^2\,M\infty^4\sin^2\beta - 4\,(M\infty^2\sin^2\beta - 1)\,(\gamma\,M\infty^2\sin^2\beta + 1)] / [[2\,\gamma\,M\infty^2\sin^2\beta - (\gamma - 1)]\,[(\gamma - 1)\,M\infty^2\sin^2\beta + 2]].$$

References

1. J.Katz and A.Plotkin, “Low Speed Aerodynamics”, second edition, Cambridge University Press, 2001, sec. 3.12, pp 67-59
2. C.P.Hoult, ”Modified Newtonian Drag Estimates”, RST memo, 15 November 2009
3. Ames Research Staff, “Equations, Tables, and Charts for Compressible Flow”,N.A.C.A. Report 1135, 1953